Scientific verification

Nuclear Reaction Lab v1.0.0 / review release

Executed evidence, independently computable benchmarks, documented data and stated limitations. Prepared 2026-10-14; no publication or external certification is claimed.

VALIDATION GROUP	EXECUTED	PASS	FAIL
Physics, data, units & export	124	124	0
React interface	9	9	0
Browser end-to-end	6	6	0
TOTAL	139	139	0

Reference database

18 documented records, 0 incomplete metadata. Six CODATA2022 bare masses are calculation-ready; twelve AME2020 neutral-atom records are catalog-only. No unverified electron binding correction is invented.

Checks beyond test counts

Type checking, production build, portable standalone build and DOM build checks passed. Browser tests actually downloaded PDF, CSV and PNG, exercised Arabic mobile layout, and ran the portable HTML with zero network requests. GitHub workflows were authored but not executed on GitHub.

Interpretation and outstanding limits

Complete two-body and angular mapping currently requires a stationary target. Covariance is not imported, so no joint kinematic uncertainty is propagated. Cross-section density is supplied by the user. Coulomb factors are approximate model suppression, never measured reaction probability. The app has not received external institutional scientific certification.

Why the benchmark digits are long

The following tables compare nominal-input binary64 results with exact algebra and a separate Python Decimal80 oracle that imports no application code. Their displayed digits describe arithmetic differences, not experimental accuracy. Full raw evidence and source hashes are included in the package.

Calculation evidence

Calculators 01-02 / all ten are independently checked

FIELD	01 Reaction threshold
Equation	max(0, (Mf-Mi)(Mf+Mi)/(2mA))
Data / methods source	PDG2026; exact synthetic masses; Decimal80 oracle
Reference test case	m=2,3->3,3
Independent expected	1.8333333333333 MeV
Application result	1.8333333333333 MeV
Absolute difference	0 MeV
Status / acceptance	PASS; tolerance 3.6866666666667e-09 MeV
Executed specific suite	6 assertions; boundaries, invalid inputs and independent comparisons; shared tests are additional.
Limits	Stationary target; no dynamics or Coulomb-accessibility claim.

FIELD	02 CM energy
Equation	KCM=√s-ma-mA (factorized implementation)
Data / methods source	PDG2026; exact synthetic masses; Decimal80 oracle
Reference test case	m=1,1; projectile T=1; target stationary
Independent expected	0.44948974278318 MeV
Application result	0.44948974278318 MeV
Absolute difference	0 MeV
Status / acceptance	PASS; tolerance 9.1897948556636e-10 MeV
Executed specific suite	8 assertions; boundaries, invalid inputs and independent comparisons; shared tests are additional.
Limits	Massive on-shell incident particles; moving target supported.

Calculation evidence

Calculators 03-04 / all ten are independently checked

FIELD	03 Nuclear recoil
Equation	$T_f = p_f^2/(\sqrt{(m^2+p_f^2)}+m)$
Data / methods source	PDG2026; exact 3-4-5 identity
Reference test case	m=4, initial p=0, transfer=(3,0,0)
Independent expected	1 MeV
Application result	1 MeV
Absolute difference	0 MeV
Status / acceptance	PASS; tolerance 2.02e-09 MeV
Executed specific suite	5 assertions; boundaries, invalid inputs and independent comparisons; shared tests are additional.
Limits	Specified frame and initial momentum; no impulse dynamics.

FIELD	04 Two-body kinematics
Equation	$p^* = \sqrt{\lambda(s, m_b^2, m_B^2)}/(2\sqrt{s})$, Lorentz boost
Data / methods source	PDG2026; exact equal-mass symmetry; Decimal80
Reference test case	all masses=1000; T=600; CM90°
Independent expected	300 MeV
Application result	300 MeV
Absolute difference	0 MeV
Status / acceptance	PASS; tolerance 6.0002e-07 MeV
Executed specific suite	22 assertions; boundaries, invalid inputs and independent comparisons; shared tests are additional.
Limits	Stationary target; labelled two-body products; fixed masses.

Calculation evidence

Calculators 05-06 / all ten are independently checked

FIELD	05 Angle transformation
Equation	atan2(p* sinθ*, γ(p* cosθ*+βEb*))
Data / methods source	PDG2026 Lorentz boost; independent exact inverse-root algebra
Reference test case	m=4,1 elastic; T=3; lab asin(1/8), high branch
Independent expected	41.066248911138 deg
Application result	41.066248911138 deg
Absolute difference	7.105427357601e-15 deg
Status / acceptance	PASS; tolerance 8.2152497822276e-08 deg
Executed specific suite	11 assertions; boundaries, invalid inputs and independent comparisons; shared tests are additional.
Limits	Multiple preimages retained; zero-momentum directions undefined.

FIELD	06 Outgoing detector energy
Equation	Etot√(p²+mb²)-Pp cosθ=(s+mb²-mB²)/2
Data / methods source	PDG2026; independent equal-mass lab45 closed form
Reference test case	all masses=1000; T=600; lab45°
Independent expected	260.86956521739 MeV
Application result	260.86956521739 MeV
Absolute difference	1.1368683772162e-13 MeV
Status / acceptance	PASS; tolerance 5.2175913043478e-07 MeV
Executed specific suite	5 assertions; boundaries, invalid inputs and independent comparisons; shared tests are additional.
Limits	Lab detector angle; all directed physical roots, no detector response.

Calculation evidence

Calculators 07-08 / all ten are independently checked

FIELD	07 Momentum transfer
Equation	$q=p_i-p_f$; $t=(E_i-E_f)^2-q^2$
Data / methods source	PDG2026 four-vectors; independent vector geometry
Reference test case	pi=(0,0,5), pf=(4,0,3), Ei=Ef=10
Independent expected	4.4721359549996 MeV/c
Application result	4.4721359549996 MeV/c
Absolute difference	0 MeV/c
Status / acceptance	PASS; tolerance 8.9642719099992e-09 MeV/c
Executed specific suite	5 assertions; boundaries, invalid inputs and independent comparisons; shared tests are additional.
Limits	Same frame; t requires total energies, not just kinetic energies.

FIELD	08 Momentum balance
Equation	$\Delta p=\Sigma p_{in}-\Sigma p_{out}$; scale-aware floating-point tolerance
Data / methods source	Independent vector sums; exact deliberate perturbation
Reference test case	(2,4,2) initial vs (2+1e-8,4,2) final
Independent expected	1e-08 MeV/c
Application result	9.9999999392253e-09 MeV/c
Absolute difference	6.0774709918447e-17 MeV/c
Status / acceptance	PASS; tolerance 2.000002e-11 MeV/c
Executed specific suite	5 assertions; boundaries, invalid inputs and independent comparisons; shared tests are additional.
Limits	Floating-point closure; not an experimental confidence test.

Calculation evidence

Calculators 09-10 / all ten are independently checked

FIELD	09 Differential cross-section transformation
Equation	J=D³/(γ|1+r cosθ*|); flab=ΣfCM,k Jk
Data / methods source	PDG2026 boost, independent derivative; supplied synthetic densities
Reference test case	m=4,1 elastic; T=3; lab0; supplied values1,1 mb/sr
Independent expected	38.258064516129 mb/sr
Application result	38.258064516129 mb/sr
Absolute difference	7.105427357601e-15 mb/sr
Status / acceptance	PASS; tolerance 7.6536129032258e-08 mb/sr
Executed specific suite	6 assertions; boundaries, invalid inputs and independent comparisons; shared tests are additional.
Limits	Supplied angular densities only; branch-resolved inverse; singular folds withheld.

FIELD	10 Coulomb barrier & penetration model
Equation	VC=Z1 Z2 α ħc/R; PG=exp(-2πη)
Data / methods source	CODATA2022 α; exact SI h,c,e; IAEA LEXFOR; synthetic R,m
Reference test case	Z1=Z2=1, m1=m2=1000, Erel=.01, R=5fm
Independent expected	0.28799290937205 MeV
Application result	0.28799290937205 MeV
Absolute difference	0 MeV
Status / acceptance	PASS; tolerance 5.9598581874411e-10 MeV
Executed specific suite	8 assertions; boundaries, invalid inputs and independent comparisons; shared tests are additional.
Limits	Classical contact barrier; nonrelativistic l=0 model, not reaction probability.

Sources and reproducibility

Primary source records

NIST CODATA2022 bare masses and constants

https://physics.nist.gov/cuu/Constants/Table/allascii.txt

AME2020 original masses

https://amdc.impcas.ac.cn/masstables/Ame2020/mass_1.mas20

PDG2026 Kinematics

https://pdg.lbl.gov/2026/reviews/rpp2026-rev-kinematics.pdf

IAEA LEXFOR2015/08, A.7

https://www-nds.iaea.org/nrdc/nrdc_doc/iaea-nds-0208-201508.pdf

Catchen, Husain and Zare scattering transformations

https://www-leland.stanford.edu/group/Zarelab/publinks/151.pdf

Gamow-factor limitations, Hwang et al.

https://arxiv.org/abs/2302.10102

Reproduce the results

npm ci; npm run data:validate; npm run typecheck; npm test; python3 scripts/independent_reference.py; npm run build; npm run verify:build; npx playwright install --with-deps chromium; npm run test:e2e.

Development failures were fixed

Independent review exposed low-energy cancellation and spurious roots; React tests exposed a null-input rendering error; the browser offline test exposed an embedding-string bug. Corrections and regression cases are described in TEST_REPORT.md. The final formal test run has zero failing or skipped declared tests.